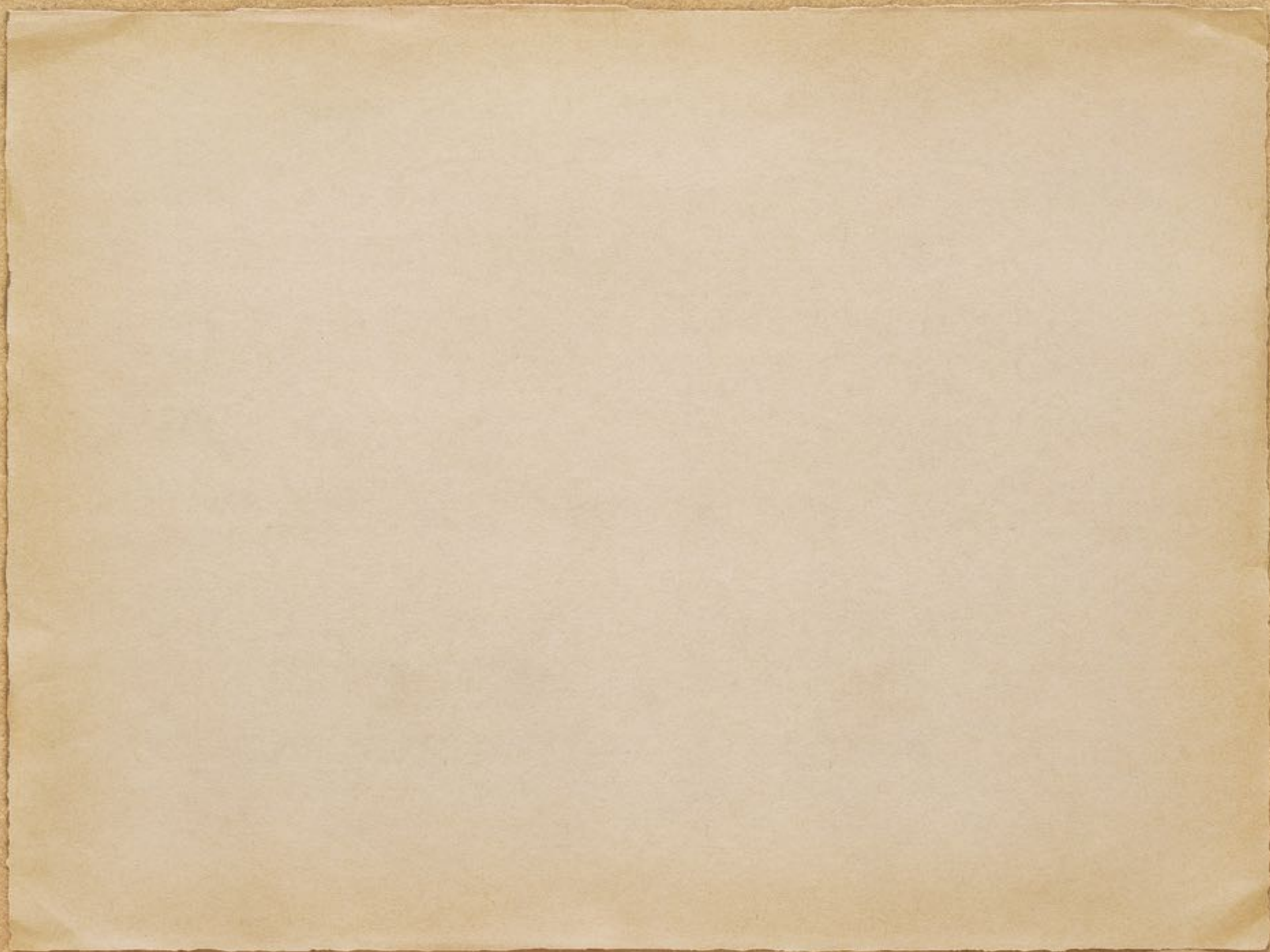
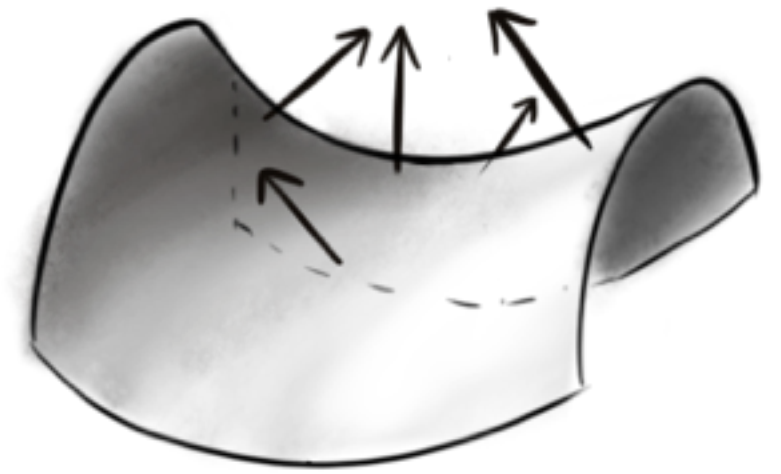
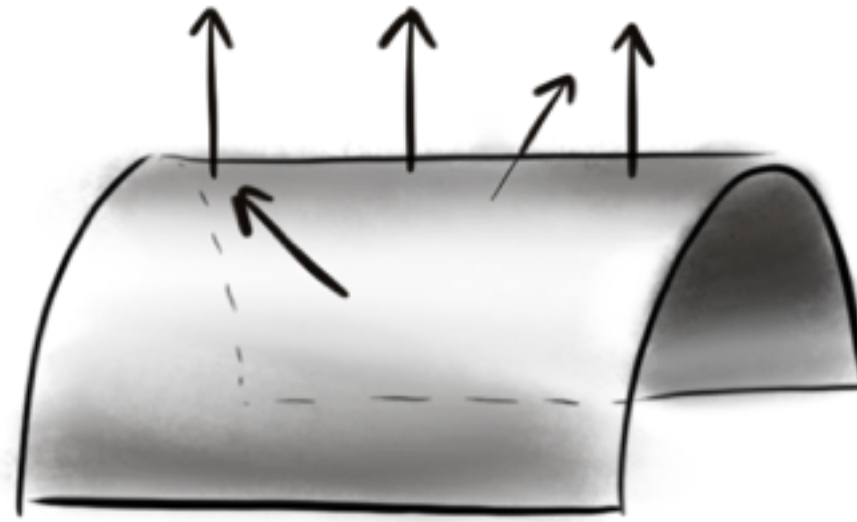




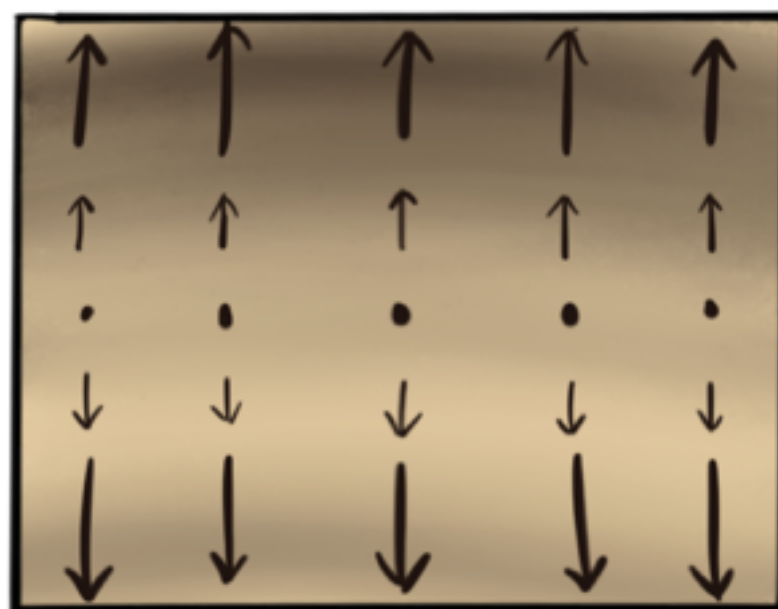
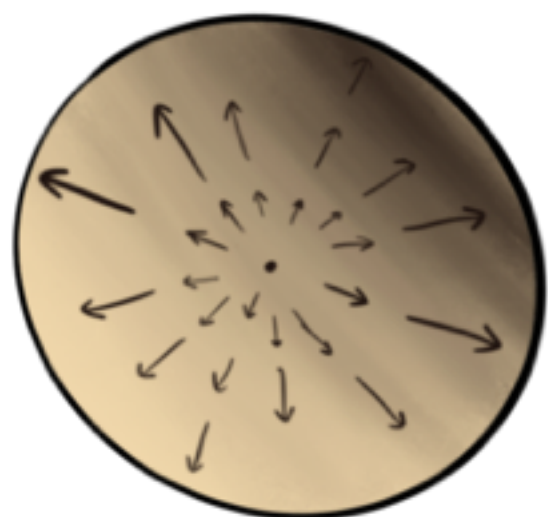


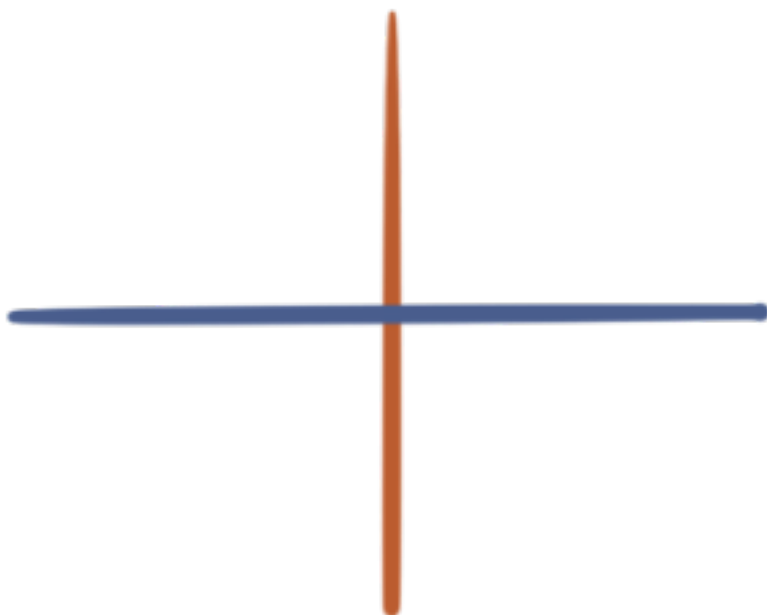
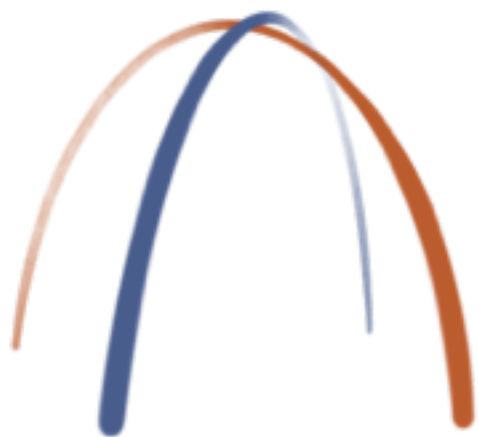


# Local structure of gauss map







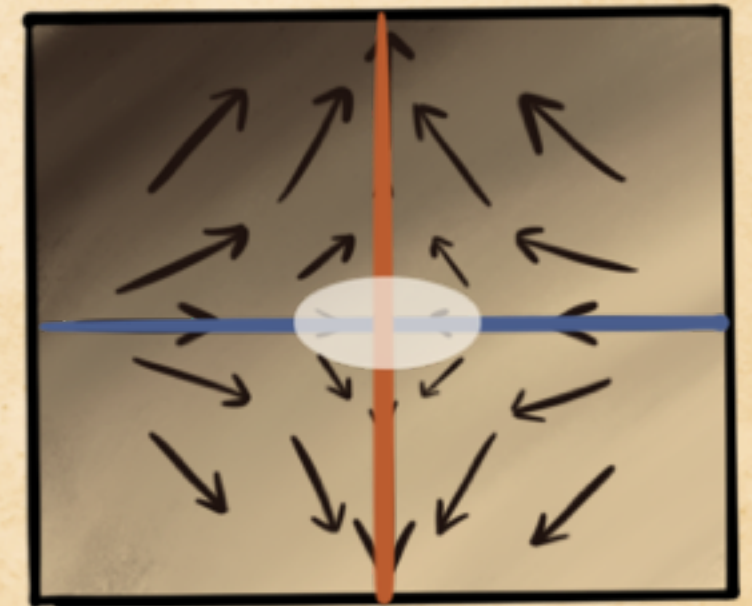
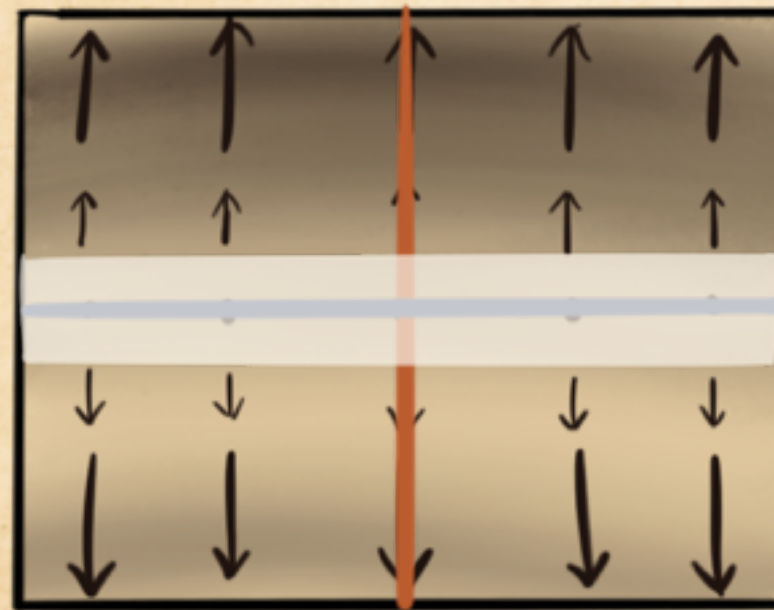
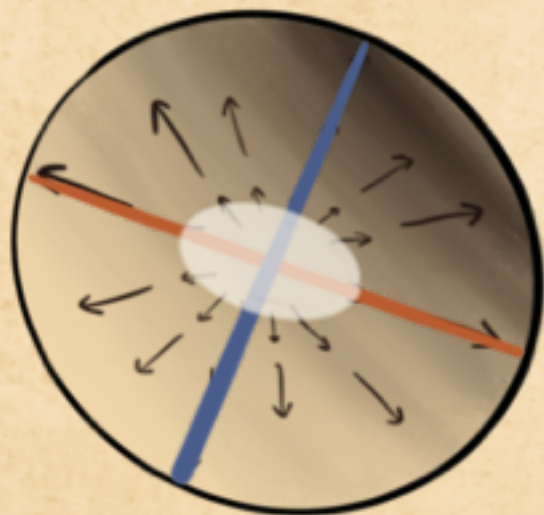
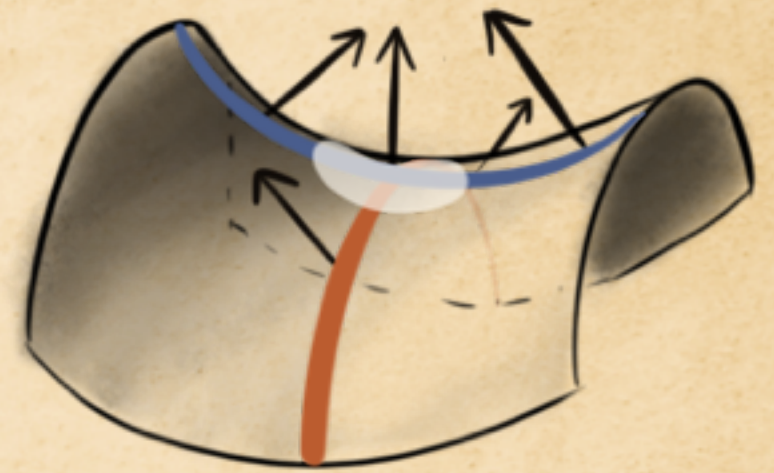
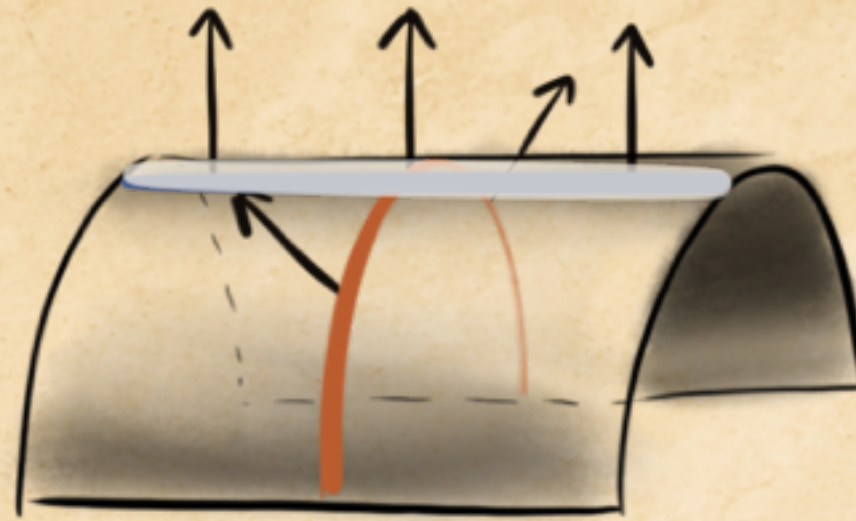




Preimage of circle  $\bigcirc \xrightarrow{G} \bigcirc$  stretches according  
to rate of change of the normal vector



# Local structure of gauss map



Preimage of circle  $\bigcirc \xrightarrow{G} \bigcirc$  stretches according to rate of change of the normal vector



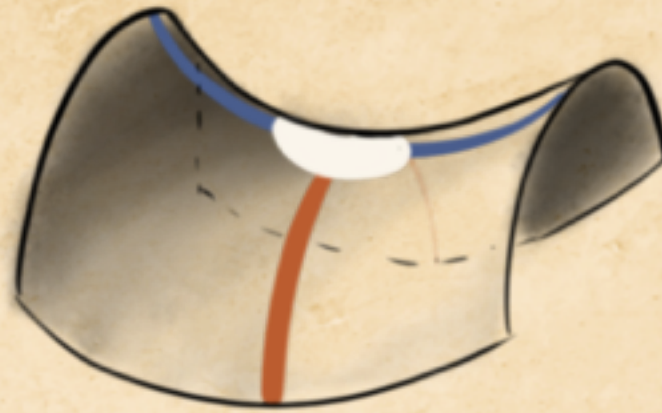
# Small highlight $\longleftrightarrow$ Large curvature



axes of ellipse

length of axis

Area of ellipse



directions of maximal/  
minimal curvature

$1/\text{Principal curvature}$

$1/\text{Gaussian curvature}$